Exercises: Products and Quotients of Vector Spaces

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*Axler, **Linear Algebra Done Right**, 4th ed., §3E*

Exercise 3

Suppose $V_1, \dots, V_m$ are vector spaces. Prove that $\mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$ are isomorphic vector spaces.

Idea. A linear map out of a product is determined by what it does on each factor separately. So we restrict T to each slot, and in the other direction we add the pieces back together.

Setup. For $k = 1, \dots, m$ define the *inclusion* $R_k \colon V_k \to V_1 \times \cdots \times V_m$ by

$$R_k(v) = (0, \dots, 0, \underbrace{v}_{k\text{th slot}}, 0, \dots, 0).$$

Each R_k is linear, since addition and scalar multiplication in the product are done one slot at a time. Note also that every element of the product splits as

$$(v_1, \dots, v_m) = R_1 v_1 + \cdots + R_m v_m. \tag{$*$}$$

Proof. Define

$$\Phi \colon \mathcal{L}(V_1 \times \cdots \times V_m, W) \to \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W), \qquad \Phi(T) = (TR_1, \dots, TR_m).$$

Each TR_k is a composition of linear maps, so $TR_k \in \mathcal{L}(V_k, W)$ and Φ is well defined. We show that Φ is an isomorphism.

Φ is linear. For $S, T \in \mathcal{L}(V_1 \times \cdots \times V_m, W)$ and $\lambda \in \mathbf{F}$, composition distributes over addition and scalar multiplication (3.8), so

$$\Phi(S+T) = ((S+T)R_1, \dots, (S+T)R_m) = (SR_1 + TR_1, \dots, SR_m + TR_m) = \Phi(S) + \Phi(T),$$

$$\Phi(\lambda T) = ((\lambda T)R_1, \dots, (\lambda T)R_m) = \lambda\,(TR_1, \dots, TR_m) = \lambda\,\Phi(T).$$

Φ is injective. Suppose $\Phi(T) = 0$, so $TR_k = 0$ for every k. Then for any $(v_1, \dots, v_m)$, by $(*)$,

$$T(v_1, \dots, v_m) = T(R_1 v_1) + \cdots + T(R_m v_m) = 0 + \cdots + 0 = 0.$$

So $T = 0$, which means $\operatorname{null} \Phi = \{0\}$ and Φ is injective (3.15).

Φ is surjective. Let $(S_1, \dots, S_m) \in \mathcal{L}(V_1, W) \times \cdots \times \mathcal{L}(V_m, W)$. Define $T \colon V_1 \times \cdots \times V_m \to W$ by

$$T(v_1, \dots, v_m) = S_1 v_1 + \cdots + S_m v_m.$$

T is linear, because

$$\begin{aligned} T((v_1,\dots,v_m)+\lambda(u_1,\dots,u_m)) &= T(v_1+\lambda u_1,\dots,v_m+\lambda u_m)\\ &= \textstyle\sum_k S_k(v_k+\lambda u_k)\\ &= \textstyle\sum_k S_k v_k + \lambda\sum_k S_k u_k\\ &= T(v_1,\dots,v_m)+\lambda\,T(u_1,\dots,u_m). \end{aligned}$$

Also, for $v \in V_k$, every slot of $R_k v$ is 0 except slot k, so $TR_k v = S_k v$. Hence $TR_k = S_k$ for each k, and $\Phi(T) = (S_1,\dots,S_m)$.

Since Φ is linear, injective, and surjective, it is an isomorphism. So $\mathcal{L}(V_1\times\cdots\times V_m, W) \cong \mathcal{L}(V_1,W)\times\cdots\times\mathcal{L}(V_m,W)$. ■

Remarks.

1. The proof never uses finite dimension, so the result holds for all vector spaces. That is why we construct an explicit isomorphism instead of comparing dimensions.

Exercise 4

> Suppose $W_1,\dots,W_m$ are vector spaces. Prove that $\mathcal{L}(V, W_1\times\cdots\times W_m)$ and $\mathcal{L}(V,W_1)\times\cdots\times\mathcal{L}(V,W_m)$ are isomorphic vector spaces.

Idea. This is the mirror image of Exercise 3. A linear map *into* a product is determined by its coordinate maps. So we read off each coordinate of T, and in the other direction we stack the pieces side by side.

Setup. For $k = 1,\dots,m$ define the *projection* $P_k\colon W_1\times\cdots\times W_m \to W_k$ by

$$P_k(w_1,\dots,w_m) = w_k.$$

Each P_k is linear, since addition and scalar multiplication in the product are done one slot at a time. Note also that every element of the product is determined by its coordinates:

$$w = (P_1 w,\dots,P_m w) \quad \text{for all } w \in W_1\times\cdots\times W_m. \tag{$**$}$$

Proof. Define

$$\Phi\colon \mathcal{L}(V,W_1\times\cdots\times W_m)\to\mathcal{L}(V,W_1)\times\cdots\times\mathcal{L}(V,W_m), \qquad \Phi(T) = (P_1T,\dots,P_mT).$$

Each P_kT is a composition of linear maps $V \xrightarrow{T} W_1\times\cdots\times W_m \xrightarrow{P_k} W_k$, so $P_kT\in\mathcal{L}(V,W_k)$ and Φ is well defined. We show that Φ is an isomorphism.

Φ is linear. For $S,T\in\mathcal{L}(V,W_1\times\cdots\times W_m)$ and $\lambda\in\mathbf{F}$, composition distributes over addition and scalar multiplication (3.8), so

$$\Phi(S+T) = (P_1(S+T),\dots,P_m(S+T)) = (P_1S+P_1T,\dots,P_mS+P_mT) = \Phi(S)+\Phi(T),$$

$$\Phi(\lambda T) = (P_1(\lambda T),\dots,P_m(\lambda T)) = \lambda\,(P_1T,\dots,P_mT) = \lambda\,\Phi(T).$$

Φ is injective. Suppose $\Phi(T)=0$, so $P_kT=0$ for every k. Then for any $v\in V$, by $(**)$,

$$Tv = (P_1Tv,\dots,P_mTv) = (0,\dots,0) = 0.$$

So $T=0$, which means $\operatorname{null}\Phi = \{0\}$ and Φ is injective (3.15).

Φ is surjective. Let $(S_1,\dots,S_m)\in\mathcal{L}(V,W_1)\times\cdots\times\mathcal{L}(V,W_m)$. Define $T\colon V\to W_1\times\cdots\times W_m$ by

$$Tv = (S_1v,\dots,S_mv).$$

T is linear, because for $u, v \in V$ and $\lambda \in \mathbf{F}$,

$$\begin{aligned} T(v + \lambda u) &= (S_1(v + \lambda u), \dots, S_m(v + \lambda u)) \\ &= (S_1 v + \lambda S_1 u, \dots, S_m v + \lambda S_m u) \\ &= (S_1 v, \dots, S_m v) + \lambda\,(S_1 u, \dots, S_m u) \\ &= Tv + \lambda\, Tu. \end{aligned}$$

Also $P_k T v = S_k v$ for every $v \in V$, so $P_k T = S_k$ for each k, and $\Phi(T) = (S_1, \dots, S_m)$.

Since Φ is linear, injective, and surjective, it is an isomorphism. So $\mathcal{L}(V, W_1 \times \cdots \times W_m) \cong \mathcal{L}(V, W_1) \times \cdots \times \mathcal{L}(V, W_m)$. ■

Remarks.

1. As in Exercise 3, nothing here uses finite dimension, so the result holds for all vector spaces.
2. Compare the two exercises: maps *out of* a product are handled with the inclusions R_k (precompose, $T \mapsto TR_k$), while maps *into* a product are handled with the projections P_k (postcompose, $T \mapsto P_k T$).